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Original Manuscript

DeepB^{3}P: A transformer-based model for identifying blood-brain barrier penetrating peptides with data augmentation using feedback GAN

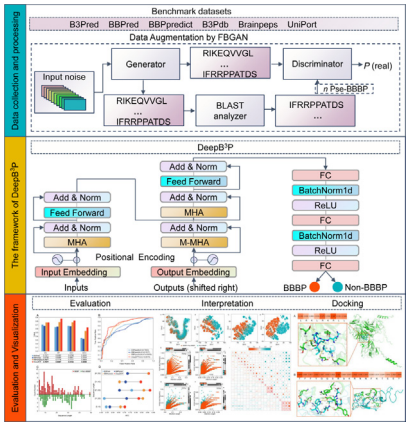
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HIGHLIGHTS

- A transformer-based deep learning model, called DeepB^{3}P, was proposed for predicting BBB penetrating peptides (BBBP).
- The FBGAN model was first utilized to synthesize analogous BBBPs to solve data imbalance issue.
- DeepB^{3}P demonstrates superior predictive performance compared to existing methodologies on benchmark datasets.
- The interpretive analysis of DeepB^{3}P contributes to the design and optimization of BBBPs.
- The analysis of attentional mechanisms in DeepB^{3}P facilitates a comprehensive understanding of interactions between BBBPs and receptors.
- The availability of the DeepB^{3}P web service facilitates the utilization of the model.

GRAPHICAL ABSTRACT



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ABSTRACT

Introduction: The blood–brain barrier (BBB) serves as a critical structural barrier and impedes the entry of most neurotherapeutic drugs into the brain. This poses substantial challenges for central nervous system (CNS) drug development, as there is a lack of efficient drug delivery technologies to overcome this obstacle. BBB penetrating peptides (BBBPs) hold promise in overcoming the BBB and facilitating the delivery of drug molecules to the brain. Therefore, precise identification of BBBPs has become a crucial step in CNS drug development. However, most computational methods are designed based on conventional models that inadequately capture the intricate interaction between BBBPs and the BBB. Moreover, the performance of these methods was further hampered by unbalanced datasets.

Objectives: This study addresses the problem of unbalanced datasets in BBBP prediction and proposes a powerful predictor for efficiently and accurately identifying BBBPs, as well as generating analogous BBBPs.

Methods: A transformer-based deep learning model, DeepB^{3}P, was proposed for predicting BBBP. The feedback generative adversarial network (FBGAN) model was employed to effectively generate analogous BBBPs, addressing data imbalance.

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Results: The FBGAN model possesses the ability to generate novel BBBP-like peptides, effectively mitigating the data imbalance in BBBP prediction. Extensive experiments on benchmarking datasets demonstrated that DeepB^{3}P outperforms other BBBP prediction models by approximately 9.09%, 4.55% and 9.41% in terms of specificity, accuracy, and Matthew's correlation coefficient, respectively. For accelerating the progress in BBBP identification and CNS drug design, the proposed DeepB^{3}P was implemented as a webserver, which is accessible at http://cbcb.cdutcm.edu.cn/deepb3p/.
Conclusion: The interpretable analyses provided by DeepB^{3}P offer valuable insights and enhance downstream analyses for BBBP identification. Moreover, the BBBP-like peptides generated by FBGAN hold potential as candidates for CNS drug development.
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Introduction

The discovery and development of novel and effective bioactive drug molecules for various central nervous system (CNS) disorders pose significant challenges [1]. In addition to the pathophysiological of CNS disorders, the intricate nature of the brain presents a substantial obstacle in pharmaceutical development. Significantly, the existence of the blood–brain barrier (BBB) presents a critical challenge due to the current lack of effective technologies for drug delivery to the brain [2,3]. The BBB acts as a crucial physiological barrier, separating blood vessels from the brain parenchyma and effectively preventing the entry of 98 % of small molecules and 100 % of large molecules into the brain [4–6]. To overcome the hurdles of drug delivery to the brain, one promising strategy is the use of BBB-penetrating peptides (BBBPs) as brain delivery molecular vectors for enhanced transport across the BBB [7,8]. Owing to their easy of synthesis, lower immunogenicity, and low toxicity, BBBPs have become vectors for delivering drugs to the CNS [9–11].

BBBPs primarily originate from neurotropic endogenous proteins, neurotoxins, certain viruses, and endogenous peptides [12]. They can also be identified through techniques such as phage display biopanning and modification [12]. Assessing the capacity of a peptide to cross the BBB and elucidating the magnitude of its permeability presents a challenge that demands the development of appropriate in vitro and in vivo methods. In an early study, Kastin and colleagues injected radiolabeled peptides into the carotid artery of mice and observed radioactivity in various brain regions, providing initial evidence that certain endogenous peptides can traverse the BBB [13,14]. Since then, an increasing number of peptides have been recognized for their ability to cross the BBB, prompting the development of multiple in vitro and in vivo methodologies to validate BBBPs. Currently, several advanced in vitro models, including the BBB specific parallel artificial membrane permeability (BBB-PAMPA), bovine brain microvascular endothelial cells (BBMECs), and induced pluripotent stem cells (iPSC)-based models, have been developed to more accurately simulate the in vivo environment and facilitate the assessment of the peptide's ability to cross the BBB [15–17]. In vivo experimental techniques include measuring the brain uptake index (BUI), employing multiple time regression (MTR) analysis, in situ brain perfusion, brain microdialysis, and quantitative radiography [18–21]. These methods yield comprehensive and detailed quantitative data on the penetration of peptides across the BBB.

Simultaneously with these advancements, researchers have made substantial progress in identifying BBBPs and constructing comprehensive databases. Although traditional biomedical measurements from in vitro and in vivo experiments are reliable, they involve significant costs and lengthy development cycles, limiting their application to large-scale data. As a result, the number of experimentally identified BBBPs remains limited, and they are dispersed across databases such as Brainpeps and B3Pdb [22,23]. With the rapid development of artificial intelligence technology, machine learning methods have been applied in numerous research fields [24,25]. Machine learning-based methods offer advantages over experimental methods in identifying high-confidence BBBPs by significantly narrowing down the search scope of candidate peptides. Several methods, such as B3Pred, BBPpred, BBPpredict and DeepB^{3}P^3, have been developed for predicting BBBPs [26–29]. Among them, B3Pred, BBPpred and BBPpredict are based on traditional machine learning techniques, employing manually designed features and a relatively smaller training dataset [26–28]. While these models offer computational convenience and partial interpretability, they have limitations in further optimization during training compared to deep learning methods. Additionally, they lack the ability to capture the intricate interactions between BBBPs and BBB. Furthermore, the limited training data may result in poor generalization ability and reduced prediction performance for novel peptides. DeepB^{3}P^3 is a deep learning-based method that employs data augmentation technique by random masking and reversing the BBBPs [29]. Although DeepB^{3}P^3 addresses the issue of data imbalance, however, the employed data augmentation method introduces an excessive amount of redundant information. Moreover, it is worth noting that the unavailability of the DeepB^{3}P^3 online poses a disadvantage to biologists.

Due to the substantial costs and labor involved in the discovery and validation of peptides, the number of peptides definitively identified as capable of crossing the BBB is limited. It is widely recognized that a lack of data and imbalanced data present challenges to training high-performing machine learning models [30]. Generative Adversarial Networks (GANs) excel in capturing the underlying data distribution and generating entirely novel instances through learned distribution, surpassing the effectiveness of oversampling or up-sampling techniques. Recently, GANs have been increasingly utilized to generate realistic biological data, such as DNA, peptide and protein sequence [31–33]. Gupta et al. initially introduced the feedback GAN (FBGAN) as a method to optimize synthetic gene sequences for desired properties, utilizing an external function analyzer in the optimization process [33]. The FBGAN comprises two main components: the Wasserstein GAN with a gradient penalty (WGAN-GP) and the function analyzer. WGAN-GP employs the Wasserstein loss formulation and incorporates a gradient norm penalty to enforce Lipschitz continuity [34]. On the other hand, the function analyzer, which can take the form of a versatile neural network or any other analytical function, fine-tunes the generated data to showcase the desired property [33]. FBGAN exhibits remarkable proficiency in generating synthetic genes that encode antimicrobial peptides and optimizing synthetic genes to attain the desired secondary structure in the resulting peptides [33].

With the increasing demand for CNS drugs due to the aging population, there is an urgent need to employ effective methodologies to enhance success rates in discovering and designing BBBPs. Additionally, the scarcity of experimentally validated BBBPs

presents significant opportunities for improving the performance of computational approaches. Therefore, in this study, we introduced DeepB^{3}P, a transformer-based deep learning model for predicting BBBPs. We employed an FBGAN-based model to augment the BBBPs, aiming to alleviate bias caused by unbalanced data and enhance the overall performance of DeepB^{3}P. Experimental results demonstrated that the FBGAN model effectively captured the inherent characteristics of BBBPs and generated novel peptides with similar properties for BBBPs. Moreover, by leveraging data augmentation with the FBGAN model, DeepB^{3}P outperformed state-of-the-art methods when evaluated on an independent testing dataset. To facilitate the use of DeepB^{3}P, we developed a user-friendly web interface for DeepB^{3}P, which can be accessed at https://cbcb.cdutcm.edu.cn/deepb3p/. The source code and datasets can be accessed at https://github.com/GreatChenLab/deepB3P.git.

Materials and methods

The workflow of the model was illustrated in Fig. 1, which consists five steps, namely data acquisition and processing (Fig. 1A, section 2.1), amino acid embedding (Fig. 1B, section 2.2), data augmentation using FBGAN (Fig. 1C, section 2.3), and the design of neural network architecture for DeepB^{3}P (Fig. 1D-E, section 2.3).

Benchmark datasets

In this study, the BBBPs used for training the models were obtained by extending the dataset published with BBPpredict [28]. We integrated BBBPs from publicly available databases and prediction tools that were not included in the original BBPpredict datasets. To generate the negative dataset (Non-BBBPs), the selection criteria from BBPpredict were applied [28]. The sequences from UniProt were filtered using keywords and exclusion criteria to ensure that they were not related to the BBB, brain, brain peptides, B3Pdb, permeation, permeability, venom, toxin, transmembrane, transport, transfer, membrane, neuro, or hemolysis [28]. Additionally, only reviewed sequences with a length range from 5 to 50 amino acids (aa) were included, and sequences with ambiguous residues were excluded. The CD-HIT clustering method was then employed to remove redundant sequences of the negative dataset with a threshold of 40 % [35]. For independent testing dataset, the positive and negative testing datasets from BBPpredict were reused. Finaly, this yielded a total of 329 BBBPs and 6,851 Non-BBBPs for training the model, and 99 of BBBPs and Non-BBBPs for testing the model. The length of peptide sequences in the datasets were ranged from 5 to 50 aa.

Amino acid encoding schemes

In our collected data, the length of BBBPs ranged from 5 to 50 aa. To handle the variable input length, we standardized the length of input peptides to 50 aa by padding the sequence with the X character at the end. For the FBGAN, we employed a one-hot encoding scheme for feature encoding. In this scheme, each amino acid was represented as a vector of length 21, where only one bit out of the 21 is set to 1. This unique representation distinguished the 20 aa and the padding character X. For example, the A was coded as [0, 1, 0, …, 0] with a length of 21. The dimension of the one-hot feature for a BBBP was (50 × 21)-D (Fig. 1B). For the DeepB^{3}P, we used an ordinal encoding scheme. In this scheme, each unique amino acid was assigned a corresponding integer value. For example, code 0 was assigned to X, code 1 was assigned to A, and so on (Fig. 1B).

The framework of FBGAN and DeepB^{3}P

Fig. 1C depicted the framework of FBGAN, which consists of a generator, a discriminator, and a feedback analyzer. The discriminator's objective is to distinguish between BBBPs and fake data (Pse-BBBPs) generated by the generator. Conversely, the generator aims to generate Pse-BBBPs that can deceive the discriminator. The discriminator was updated to maximize the score for BBBPs and minimize the score for Pse-BBBPs. Simultaneously, the generator parameters were adjusted to enhance the realism of the generated Pse-BBBPs. In our proposed framework, both the generator and discriminator consisted of three 1D convolution layers with a kernel size of 5, followed by one fully connected layer (FC). As a feedback analyzer, BLAST was exclusively employed to select sequences that demonstrate higher similarity to BBBPs compared to Non-BBBPs. These selected peptides were then incorporated into the discriminator training process.

Fig. 1D illustrated the architecture of the encoder and decoder in the transformer. The input features were processed through an embedding module that combines positional embedding with amino acid embedding to generate sequence embeddings. Subsequently, dropout technique was applied to enhance robustness. Given the position p in the sequence, the positional embedding is encoded as a k-dimensional vector, and the value of the i-th element is denoted as $PE(p)_i$, which can be calculated as follows:

$$\begin{cases} \mathrm{PE}(p)_{2i} = \sin(p/10,000^{2i/d_x}) \\ \mathrm{PE}(p)_{2i+1} = \cos(p/10,000^{2i/d_x}) \end{cases} \quad (1)$$

where $2i$ represents the even dimensions and $2i + 1$ stands for the odd ones. This position embedding method can reflect not only the absolute position information of the amino acid but also the relative position information.

The multi-head self-attention mechanism serves as a crucial module within the decoder and encoder architecture. The attention mechanism enables the model to focus on important information while reducing the influence of unimportant information within a large dataset. The multi-head self-attention layer had three inputs, the keys (K), the values (V) and the queries (Q). The attention was calculated as follows:

$$\mathrm{attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

where T represents the transpose operation of a matrix, and the parameter d_k is primarily utilized to adjust the dimensionality of the computation result.

The MLP presented in Fig. 1E comprising three FC layers, each integrated with batch normalization and ReLU activation functions.

The training processes

In the training process of FBGAN, the generator and discriminator were trained in a ratio of 1:5, meaning that for every training step of the generator, the discriminator was trained for five steps. The Adam algorithm was applied as the optimizer for the model, with the learning rate of 1e-4. The batch size for FBGAN was set to 64, and the training process consisted of 1,000 epochs. For every 100 epochs, the 6,522 Pse-BBBPs designed by the generator were used to train the DeepB^{3}P, which was then evaluated using the independent testing dataset.

In DeepB^{3}P, the embedding size for both position and token was set to 512. The feedforward dimension was specified as 16. The dimensions of K, V, and Q were set to 32, and the encoder and decoder in transformer architecture comprised 2 heads. DeepB^{3}P

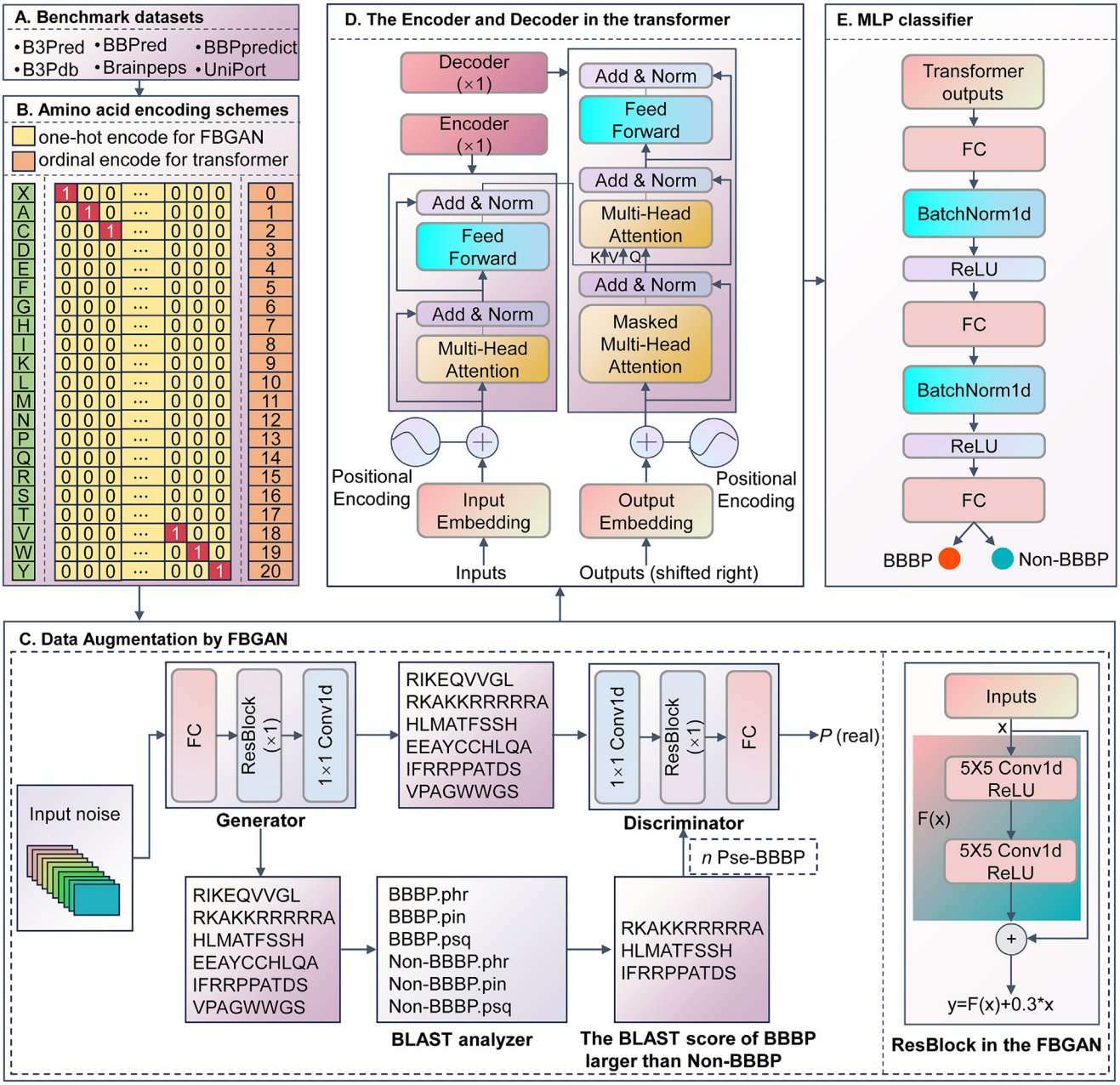


Fig. 1. The workflow of the proposed DeepB^{3}P construction. (A) main sources of datasets, (B) different amino acid embedding for FBGAN and DeepB^{3}P, (C) the architecture of FBGAN, (D) the encoder and decoder architecture in the transformer, (E) Multilayer Perceptron (MLP) architecture.

adopted the same learning rate and optimization function as FBGAN. To prevent overfitting, DeepB^{3}P was trained using a combination of five-fold cross-validation and early stopping technique. The batch size for DeepB^{3}P was set to 64, and the training process involved 200 epochs for each fold in cross-validation. By doing so, five predictive models were obtained. The output of DeepB^{3}P was the average of the values generated by these five models.

The model training and testing process was performed on a CentOS Linux system. The CPU was an Intel(R) Xeon(R) Gold 62336Y CPU @ 2.40 GHz, which had 48 logical CPUs. The GPU was the NVIDIA A100, which had a memory capacity of 80G.

Performance evaluation metrics

In this study, the performance of the proposed model was evaluated using several widely employed assessment metrics, namely Matthew's correlation coefficient (*MCC*), sensitivity (*SN*), specificity (*SP*), and accuracy (*ACC*), all of which were defined in Equation (3).

$$\begin{cases} SN = \frac{TP}{TP+FN} \\ SP = \frac{TN}{TN+FP} \\ ACC = \frac{TP+TN}{TP+TN+FN+FP} \\ MCC = \frac{TP*TN-FP*FN}{\sqrt{(TN+FP)(TN+FN)(TP+FP)(TP+FN)}} \end{cases} \quad (3)$$

where *TP*/*TN* and *FN*/*FP* represent the counts of correctly and incorrectly predicted BBBPs/Non-BBBPs, respectively. Additionally, the model's performance was also evaluated using the area under the receiver operating characteristic curve (AUC). The *AUC* values range from 0.5 and 1, with a higher score indicating superior model performance.

Result and discussion

Comparison of DeepB^{3}P with existing methods

To validate the effectiveness of DeepB^{3}P, we compared it with three baseline methods, namely B3Pred, BBPpred, and BBPpredict on the independent testing dataset. The results clearly demonstrated that our proposed method outperforms all other methods in terms of the four metrics (Fig. 2A). When compared to BBPpredict, our model exhibited competitive performance in terms of *SN*. However, DeepB^{3}P surpassed BBPpredict in terms of *SP*, *ACC*, and *MCC* by approximately 9.09 %, 4.55 %, and 9.41 %, respectively. Moreover, the DeepB^{3}P obtained an *AUC* of 0.858, surpassing all other methods (Fig. 2B).

It is important to note that not all methods are suitable for peptides with lengths ranging from 5 to 50 aa. For example, B3Pred was designed to support peptides with lengths ranging from 5 to 30 aa only. In our testing dataset, sequences ranging from 5 to 30 aa accounted for 75.25 % of the entire dataset (Fig. 2C, Fig. S1). Among the BBBPs, about 81.82 % of the peptides fall within this length range, while for the Non-BBBPs, the percentage was approximately 68.69 % (Fig. 2C, Fig. S1). To assess the prediction performance of different methods for sequences with varying lengths, we constructed five sub-datasets by partitioning the testing dataset based on length variations. The *ACC* of the methods on these sub-datasets was presented in Fig. 2D. The results indicated the effectiveness and superiority of DeepB^{3}P over other methods for peptides of all lengths, except for BBPpredict within the length range of 11 to 20 aa. While BBPpredict exhibited competitive performance on this sub-dataset, its performance was the lowest on sub-dataset with the length of peptide ranging from 31 to 40 aa. These results demonstrated that DeepB^{3}P has the high stability and superior performance compared to other methods.

Contribution of data augmentation using FBGAN

The FBGAN training process consisted of 1,000 epochs. Every 100 epoches, both the 6,522 sequences generated by the generator (Pse-BBBPs) and the 329 of BBBPs were used as positive dataset to train our proposed model. The model was then evaluated using the testing dataset. As shown in Fig. 3A, the *ACC* of DeepB^{3}P, trained using the Pse-BBBPs, improved with increasing training steps and stabilized around 800 epochs. Except for *SN*, the other three evaluation metrics reached their peak values around 1,000 epochs, while *SN* reached its peak after 800 training epochs.

To evaluate whether the trained FBGAN model has successfully learned to generate peptides with similar properties to BBBPs, we employ BBPpredict to assess the Pse-BBBPs generated at intervals of 100 epochs. The results showed that for each round of gnerated Pse-BBBPs, the predicted precentage of BBBP ranged from over

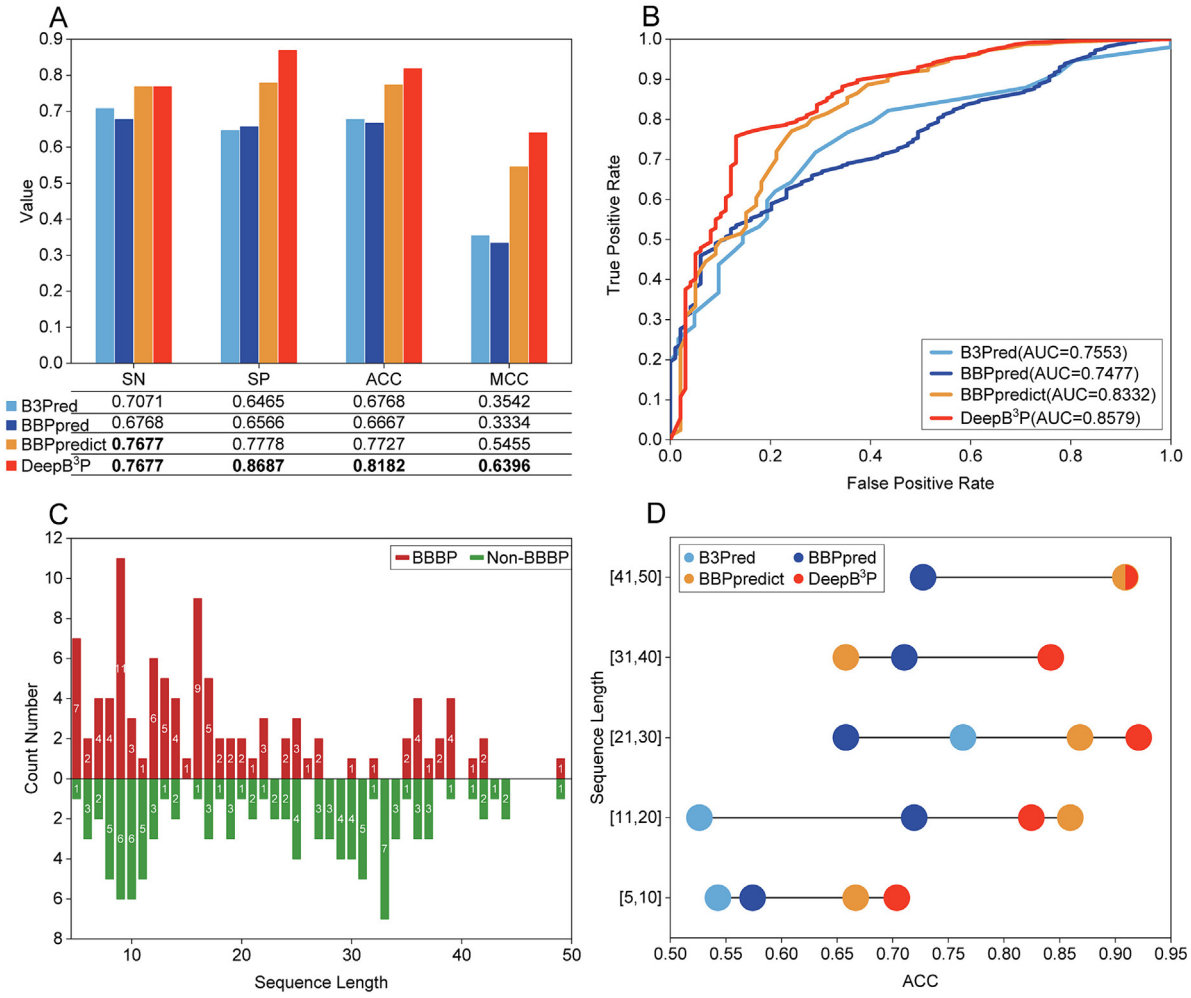


Fig. 2. Performance comparison of DeepB^{3}P with existing methods based on the independent testing dataset. (A) Performance of different methods, and the best results are marked in bold. (B) ROC curves of the four methods. (C) Peptide length distribution for BBBP and Non-BBBP samples in the independent testing dataset. (D) ACC of various methods assessed on sub-datasets divided by different length intervals.

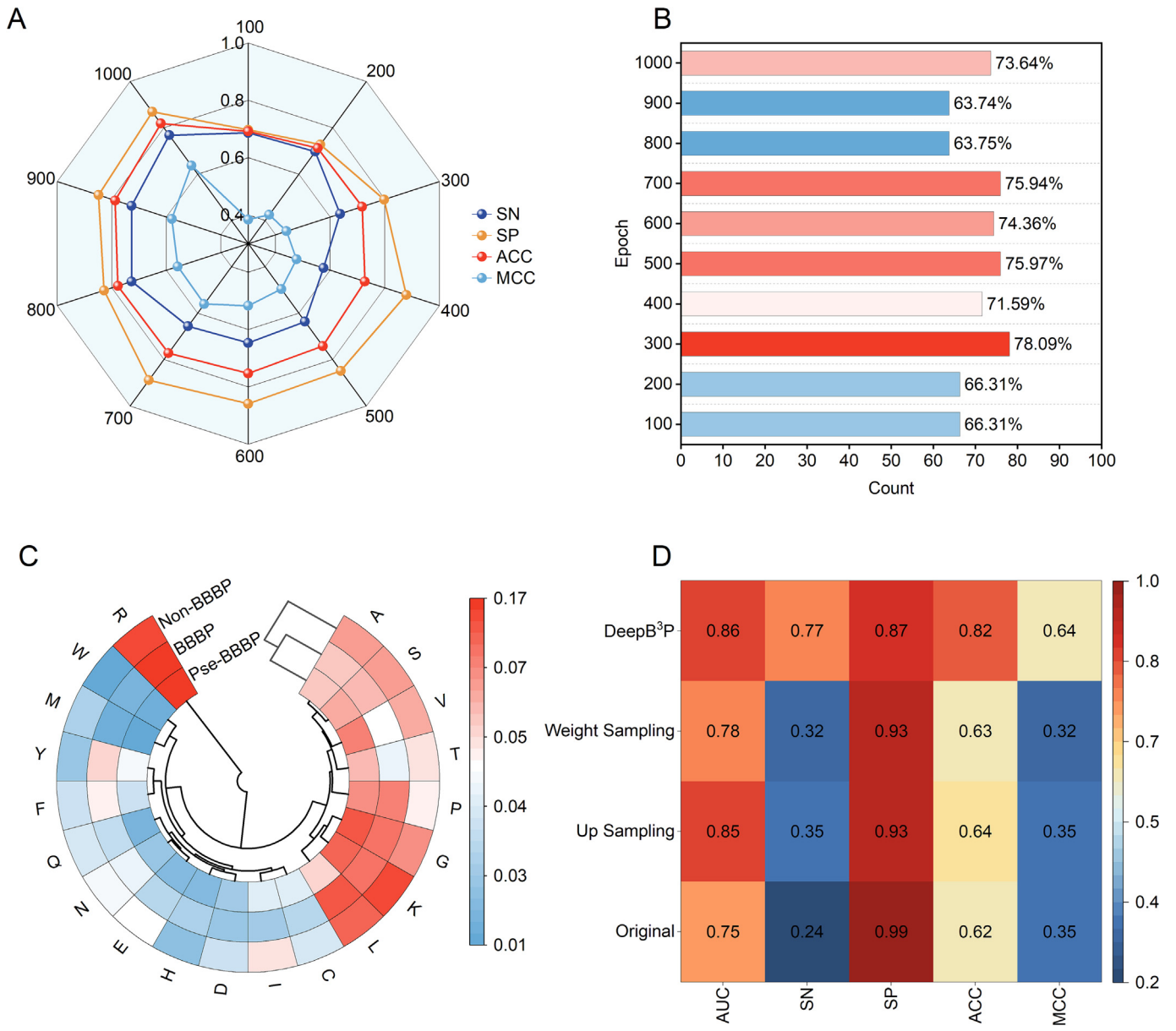


Fig. 3. Contribution analysis of data augmentation using FBGAN (A) Performance of DeepB^{3}P on testing dataset that trained with different Pse-BBBPs which geerated by FBGAN every 100 epochs. (B) Proportion of Pse-BBBPs classified as BBBPs by BBPpredict. (C) Amino acid frequencies in BBBP, Non-BBBP and Pse-BBBP generated by FBGAN at the 1000th epoch. (D) Performance of models by different data augmentation methods as well as original data.

63 % to 78 % (Fig. 3B). In the case of the 1,000th epoch, more than 73 % of the Pse-BBBPs were classified as BBBPs by BBPredict (Fig. 3B). Furthermore, we compared the amino acid frequency distributions among BBBPs, Non-BBBPs and Pse-BBBPs generated in the 1,000th epoch. The comparison revealed that the amino acid frequency distribution of Pse-BBBPs exhibited a greater similarity to that of BBBPs (Fig. 3C). These findings demonstrated that FBGAN effectively captures the frequency distribution characteristics of amino acid in BBBPs, enabling it to generate peptides that possess the remarkable ability to confound state-of-the-art prediction methods.

Additionaly, we further investigated the effectiveness of the FBGAN model. A comparative analysis was performed to evaluate the impact of different data augmentation methods, including up-sampling and weighted sampling, on the performance of DeepB^{3}P. As shown in Fig. 3D, compared to the original imbalanced dataset, the models exhibited a slight improvement in metrics such as *AUC*, *ACC* and *SN* when trained on the up sampling and weighted sampling datasets. However, the observed improvement was relatively modest. This colud be attributed to the fact that up sampling or weighted sampling methods may disrupt the dependencies among features and limit the ability of the model to discover feature correlations. On the other hand, Fig. 3D clearly demonstrated the significant positive impact of the Pse-BBBPs generated by FBGAN on the performance of DeepB^{3}P, particularly in terms of *SN*, which exhibited a remarkable improvement of over 50 %. In summary, the FBGAN model exhibited the capability to adeptly acquire the distinctive characteristics of BBBP, and the synthesized peptides derived from it can significantly enhance the efficacy of our proposed model in distinguishing BBBPs from Non-BBBPs.

Interpretability analysis

In order to examine the efficacy of the DeepB^{3}P architecture, we extracted the intermediate layer outputs of the model at different stages and mapped them onto a two-dimensional space for clustering using t-SNE. As depicted in Fig. 4, it is evident that the initial features were randomly distributed (Fig. 4A), whereas a more pronounced clustering effect becomes apparent after the encoder layer (Fig. 4B). The decoder layer additionally retained the crucial classification features, resulting in distinct clusters for positive samples (BBBPs add Pse-BBBPs) and Non-BBBPs (Fig. 4C). Ultimately, accurate identification was accomplished through the classification layer (Fig. 4D).

In addition, we investigated the relationship between different amino acid frequencies and attention weights, as shown in Fig. 4E-I and Fig. S2. Generally, there was a positive correlation between the frequency of amino acid occurrences and the attention weights. Among the positive samples, the amino acid R emerged with the highest frequency of occurrence and demonstrated a greater attention weight in comparison to Non-BBBPs (Table S1, Fig. 4E). It suggested that an increase in the content of R in the peptide design corresponds to an increased probability of crossing the BBB. Interestingly, there was a substantial disparity in the frequency of K between positive samples and Non-BBBPs, but the disparity in attention weights was not statistically significant (Fig. 4F). This observation implied that K plays a non-discriminatory role in the prediction of BBBPs by DeepB^{3}P. Eleven amino acids, namely A, N, H, K, Q, L, I, D, S, M, and E, exhibited a higher frequency in Non-BBBPs compared to positive samples (Fig. 4G, Fig. S2). The low attention weight of E in BBBPs was consistent with previous work indicating its neurotoxicity, highlighting the need for strict control over its concentration in the brain [36]. It is worth mentioning that G, as the amino acid with the smallest molecular mass, exhibited a higher frequency of occurrence and attention weights in the positive samples compared to Non-BBBPs (Fig. 4H). This finding suggested that a higher content of G enhances the ability of peptides to penetrate the BBB. The correlations of amino acids in BBBPs were visualized in Fig. 4I. It is noted that R exhibited a stronger correlation with K compared to its correlation with other amino acids (Fig. 4I). This correlation was effectively captured and enhanced by the FBGAN model in the Pse-BBBPs (Fig. S3A). In addition, E and N, as well as F and Y, also exhibited a strong correlation in the positive samples (Fig. 4I), while this phenomenon was not observed in the Non-BBBPs (Fig. S3B). These findings provided valuable insights for the design and optimization of peptides with enhanced BBB penetration capabilities.

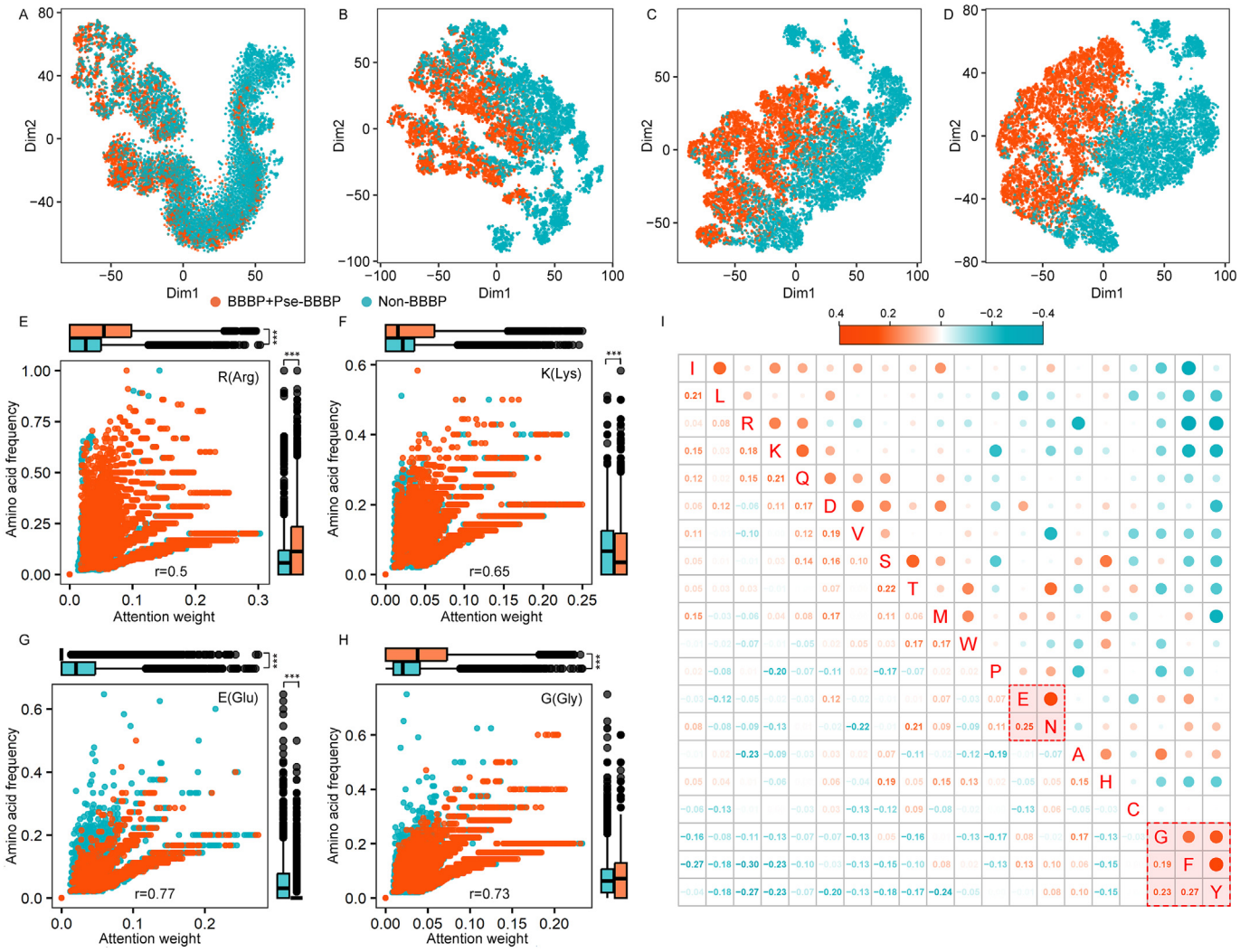


Fig. 4. Interpretability analysis for the DeepB^{3}P. (A) t-SNE visualization of the initial features. (B)-(D) t-SNE visualization of the features acquired from the transformer decoder, encoder and the penultimate layer of DeepB^{3}P, respectively. (E)-(H) frequency distribution versus attention weight distribution of amino acid in BBBPs and Non-BBBPs. (I) The amino acid correlation frequency in BBBPs.

Some BBBPs penetrate the BBB via a mechanism known as receptor-mediated transcytosis. For instance, Apolipoprotein E (ApoE) utilizes the low-density lipoprotein receptor (LDLR), and Angiopep-2 (ANG-2) interacts with the low-density lipoprotein receptor-related proteins 1 (LRP-1) [10,11]. To evaluate the advancement brought by DeepB^{3}P, we performed an analysis to investigate the intricate interplay between BBBPs and these receptors. Initially, we generated the 3D structures of ApoE and ANG-2 using the PEP-FOLD4 software [37]. The 3D structure of LDLR was predicted using AlphaFold2 [38], and the 3D structure of LRP-1 was retrieved from the Protein Data Bank website (PDB ID: 2FYL) [39]. Subsequently, we performed molecular docking using AutoDock Vina [40] and visualized the results using PyMOL [41]. The binding energies of ApoE with LDLR and ANG-2 with LRP-1 were −5.93 and −4.88 kcal/mol, respectively (Table S2). The docking results were depicted in Fig. 5. In the case of ApoE, residues L1, R2, R5, K6, R7, and L8, which were part of the active site, were found to participate in hydrogen bonding with the LDLR receptor. These residues also exhibited significant attention weights as indicated by DeepB^{3}P (Fig. 5A). Similarly, for ANG-2, the model assigned higher attentional weights to Y4, R8, N12, F14, E18, and Y19, which were also the active site residues involved in hydrogen bonding with the LRP-1 receptor (Fig. 5B). These results suggested that the attention weights served as a valuable reference for analyzing the intricate interactions between BBBPs and the specific receptors.

Furthermore, mutations were introduced at amino acid sites displaying higher attention weights in two BBBPs by substituting them with E. Afterward, we conducted an in-depth investigation into the interactions between these mutated peptides and their respective receptors. The docking results are shown in Table S3 and Fig. S4. Both mutated peptides, ANG-2-mut and ApoE-mut, exhibited over 0.5 sequence similarity to the original peptides, respectively (Table S3). The predictions of DeepB^{3}P suggested that the mutated peptides had a high probability of being unable to penetrate the BBB. Moreover, the docking simulations revealed increased binding energies for ApoE-mut with the LDLR receptor and for ANG-2-mut with the LRP-1 receptor (Table S3). Notably, the DeepB^{3}P model assigned high attention weight values to the active sites of the docked peptides (Fig. S4), further validating our model's capability to identify crucial interaction sites between peptide sequences and receptors.

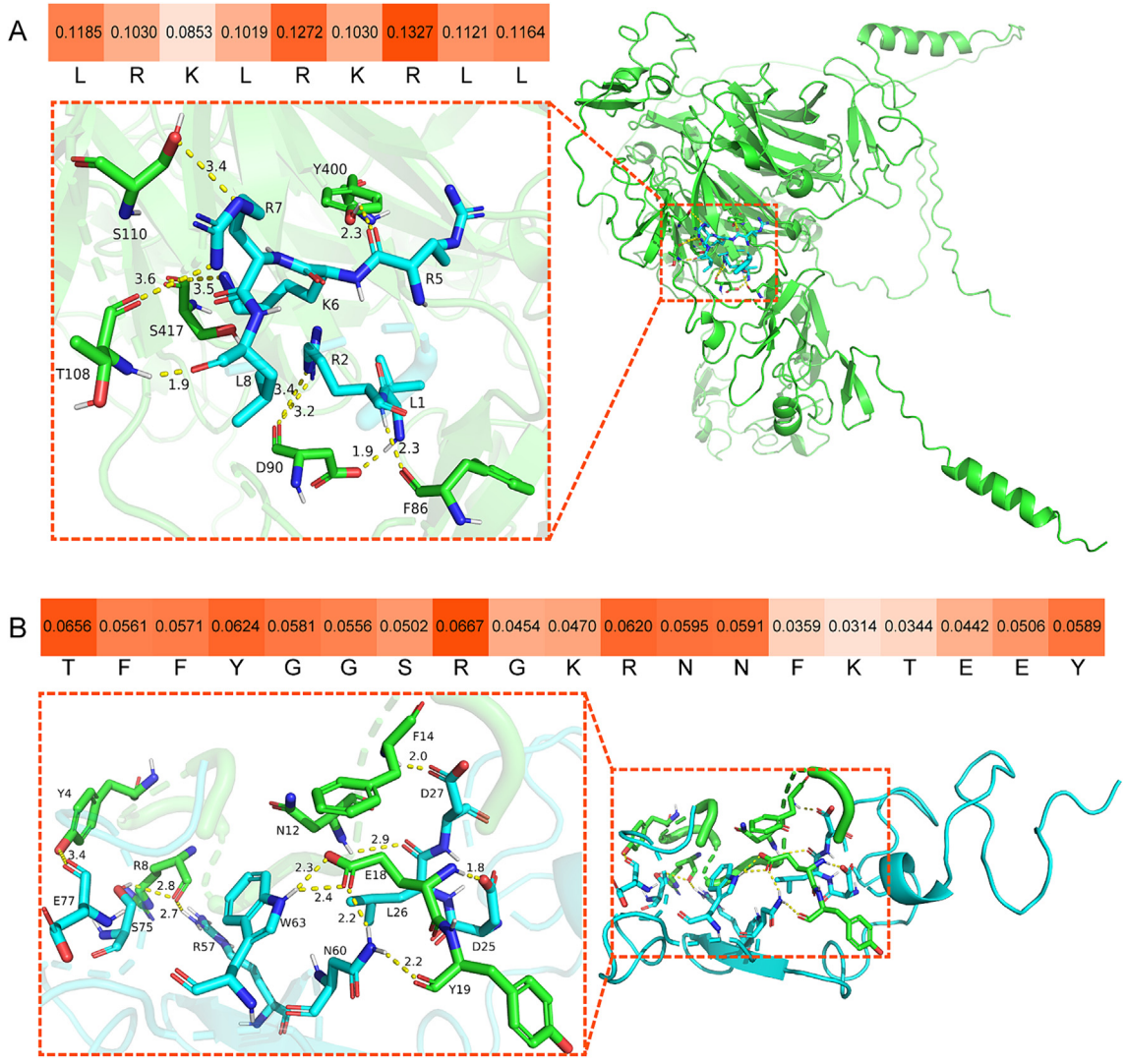


Fig. 5. Molecular docking results obtained by AutoDock vina together with the attention weights provided by DeepB^{3}P. (A) ApoE and its receptor LDLR, (B) ANG-2 and its receptor LRP-1. The heatmap presents the attention weights for each amino acid in the peptide sequences obtained from DeepB^{3}P.

Webserver implementation

To facilitate the identification and design of BBBPs, we have developed the DeepB^{3}P model as a web server, which is accessible at https://cbcb.cdutcm.edu.cn/deepb3p/.

Our webserver allows users to input a peptide sequence for classification as either BBBP or Non-BBBP. Upon submission, the server promptly determines the classification and provides the probability score indicating the likelihood of the peptide sequence falling into each category. Furthermore, users are presented with a plot depicting attention weights for further analysis. For sequences classified as Non-BBBP, our web server offers a unique functionality to mutate specific amino acids with high attention weight values. This mutation process aims to maintain the essential characteristics of the original peptide while increasing the likelihood of classification as a BBBP by the DeepB^{3}P model.

In addition to classification, our webserver includes a generator function powered by the FBGAN model trained in this study. Users can specify parameters including the number of peptides to be designed, the threshold for BBBP prediction by DeepB^{3}P, a random seed, peptide length, the amino acid composition of the peptide. Based on these inputs, the server generates a batch of candidate peptides tailored to mimic BBBPs, offering valuable insights for further research and experimentation.

Conclusion

In this study, we introduced DeepB^{3}P, a deep learning-based method for predicting BBBP. DeepB^{3}P adapted a transformer-based architecture, leveraging decoder and encoder, to capture both contextual and positional information among residues in peptide sequences. This design enables the model to effectively learn the complex relationships and dependencies within the sequences. To address the challenge of data imbalance, we incorporated an FBGAN model, which combines the power of GAN model with the feedback method to generate novel BBBP-like peptides. By iteratively training the generator and discriminator networks and incorporating feedback from the discriminator, the FBGAN model learns to generate peptides that have a higher likelihood of being recognized as BBBPs by the DeepB^{3}P model. This integrated approach effectively tackles the problem of data imbalance in BBBP prediction tasks and leads to improved performance (Fig. 3D).

To demonstrate the effectiveness of the DeepB^{3}P, we performed a comparative analysis against the state-of-the-art methods using an independent testing dataset. Additionally, we compared DeepB^{3}P with other prediction methods on sub-datasets divided into different length intervals. The results consistently showed that DeepB^{3}P outperformed these models, indicating its superior predictive performance. Simultaneously, the interpretability analysis provided valuable insights into the internal working of the DeepB^3-P, further enhances the rationality of the model. The correlation analysis of amino acid frequency and attention weight, along with the results of molecular docking, suggested that DeepB^{3}P possesses biological analytical capabilities. These findings indicated that DeepB^{3}P can provide guidance to biologists in conducting specific targeted experiments on peptides.

Furthermore, the DeepB^{3}P online server offered comprehensive prediction results and attention weights. The incorporation of FBGAN enabled biologists to generate a wide range of candidate BBBPs, providing a foundation for subsequent experimental investigations. The webserver can be conveniently accessed at https://cbcb.cdutcm.edu.cn/deepb3p/.

In summary, DeepB^{3}P is a highly competitive tool for BBBP prediction, demonstrating remarkable efficiency and accuracy, and providing insights into the field of BBBP prediction. The FBGAN model employed in this study represents a novel method for handling the prediction task of unbalanced data, particularly beneficial for small sample datasets. Additionally, the FBGAN model's capability to generate novel peptides holds immense potential for the development of innovative protein drugs, particularly for the CNS. The developed DeepB^{3}P webserver holds the potential to facilitate the researchers in achieving more effective prediction outcomes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jare.2024.08.002.

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